

Arm	Method	MSE@5 ↓	MSE@10 ↓	Gain@5 (%)	Gain@10 (%)
Slow	Framewise	0.15848	0.21626	reference	reference
Slow	Constant dynamics	0.15841	0.21605	+0.04	+0.09
Slow	ShiftWM (ours)	0.15802	0.21556	+0.29	+0.32
Slow	Action-free	0.15902	0.21564	-0.34	+0.28
Decay	Framewise	0.15907	0.21689	reference	reference
Decay	Constant dynamics	0.15899	0.21673	+0.05	+0.07
Decay	ShiftWM (ours)	0.15839	0.21948	+0.43	-1.20
Decay	Action-free	0.15976	0.21742	-0.44	-0.25
Compact	Framewise	0.15964	0.21925	reference	reference
Compact	Constant dynamics	0.15963	0.21923	+0.01	+0.01
Compact	ShiftWM (ours)	0.15888	0.22054	+0.48	-0.59
Compact	Action-free	0.16019	0.21877	-0.34	+0.22

Table 1: **Complete matched generalization development study.** All 36 runs completed 30 epochs on original training data. Entries are three-seed, equal-episode validation means; MSE uses fixed train-standardized feature coordinates. Five blocks is the primary development horizon; ten blocks is extrapolation. Slow reduces learning rate, Decay increases weight decay, and Compact jointly reduces predictor/context capacity. Gains compare each method to Framewise within the same arm. **Bold green** identifies positive point reductions, not statistical significance. Every arm and method is retained; these are validation results, not an additional held-out test.

Arm	Method	Δ MSE	Gain (%)	Paired 95% interval
Slow	Constant dynamics	-0.000065	+0.04	[-0.000111, -0.000015]
Slow	ShiftWM (ours)	-0.000458	+0.29	[-0.000671, -0.000283]
Slow	Action-free	+0.000539	-0.34	[-0.000062, +0.001162]
Decay	Constant dynamics	-0.000073	+0.05	[-0.000105, -0.000045]
Decay	ShiftWM (ours)	-0.000676	+0.43	[-0.001535, -0.000053]
Decay	Action-free	+0.000696	-0.44	[+0.000208, +0.001192]
Compact	Constant dynamics	-0.000017	+0.01	[-0.000072, +0.000039]
Compact	ShiftWM (ours)	-0.000759	+0.48	[-0.001353, -0.000278]
Compact	Action-free	+0.000543	-0.34	[+0.000029, +0.001149]

Table 2: **Matched validation uncertainty at 5 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are $100(\text{Framewise} - \text{method})/\text{Framewise}$. Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

Arm	Method	Δ MSE	Gain (%)	Paired 95% interval
Slow	Constant dynamics	-0.000201	+0.09	[-0.000322, -0.000086]
Slow	ShiftWM (ours)	-0.000698	+0.32	[-0.001144, -0.000339]
Slow	Action-free	-0.000610	+0.28	[-0.002089, +0.000888]
Decay	Constant dynamics	-0.000157	+0.07	[-0.000241, -0.000079]
Decay	ShiftWM (ours)	+0.002593	-1.20	[-0.000748, +0.005568]
Decay	Action-free	+0.000532	-0.25	[-0.000728, +0.001798]
Compact	Constant dynamics	-0.000017	+0.01	[-0.000194, +0.000153]
Compact	ShiftWM (ours)	+0.001291	-0.59	[-0.000945, +0.003425]
Compact	Action-free	-0.000483	+0.22	[-0.002460, +0.001390]

Table 3: **Matched validation uncertainty at 10 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are $100(\text{Framewise} - \text{method})/\text{Framewise}$. Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

Arm	Method	Best epochs	Total (M)	Trainable (M)	Seconds/run
Slow	Framewise	3/3/3	3.763	3.469	151 [149, 219]
Slow	Constant dynamics	3/3/3	3.763	3.714	159 [159, 234]
Slow	ShiftWM (ours)	3/3/3	3.763	3.714	161 [161, 236]
Slow	Action-free	3/3/3	3.763	3.312	141 [140, 213]
Decay	Framewise	1/1/1	3.763	3.469	148 [146, 218]
Decay	Constant dynamics	1/1/1	3.763	3.714	158 [156, 238]
Decay	ShiftWM (ours)	2/2/1	3.763	3.714	161 [158, 240]
Decay	Action-free	1/1/1	3.763	3.312	140 [138, 213]
Compact	Framewise	2/2/2	0.841	0.763	100 [99, 159]
Compact	Constant dynamics	2/2/2	0.841	0.828	110 [110, 175]
Compact	ShiftWM (ours)	2/3/3	0.841	0.828	112 [111, 173]
Compact	Action-free	2/1/2	0.841	0.722	93 [93, 149]

Table 4: **Reusable checkpoints and computation.** Best epochs are ordered by seeds 0/1/2 and selected by the shared all-five-query validation objective after full 30-epoch training. Parameter counts include frozen parameters; trainable counts exclude inactive modules. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, validation and CPU reload verification; prior interrupted work, if any, is excluded. Different GPU partitions are used, so these times are operational evidence, not a hardware-matched speed comparison. All 36 selected packages pass strict CPU reload parity.